



Islamic University of Technology
Department of Computer Science and Engineering

Lab 7: Shortest Path

CSE 4404: Algorithms Lab
Summer 2024-25

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Task A. Minimum Spanning Tree

Time Limit: 1 second | Memory Limit: 512 MB

Given a connected, undirected network of n nodes and m damaged bidirectional links, you need to find the minimum cost to restore connectivity. Each link connects two distinct nodes and has a specific repair cost. You must select a subset of links to repair such that every node can communicate with every other node, directly or indirectly, while minimizing the total repair cost.

Input Format

The first line contains two integers n and m ($1 \leq n \leq 10^5$, $1 \leq m \leq 2 \cdot 10^5$) — the number of nodes and the number of links.

Then, m lines follow, each containing three integers a , b , and c ($1 \leq a, b \leq n$, $1 \leq c \leq 10^9$) — indicating a bidirectional link between node a and node b with a repair cost of c .

There is at most one link between any two nodes.

Output Format

Print one integer — the minimum total cost to restore connectivity between all nodes.

If it is impossible to connect all nodes, print **IMPOSSIBLE**.

Examples

Sample Input	Sample Output
5 6 1 2 3 2 3 5 2 4 2 3 4 8 5 1 7 5 4 4	14

Notes

For this task, make 2 submissions with 2 different minimum spanning tree algorithms:

1. Prim's Algorithm
2. Kruskal's Algorithm

Task B. Urgent Cables

Time Limit: 2 seconds | *Memory Limit: 512 MB*

Following the cable disaster of July 4202, each damaged cable has been assigned an *urgency rating* by the engineering team. For a cable connecting data centers a and b with repair cost c , the urgency rating is defined as:

$$p = c \cdot (a + b)$$

The repair crew must work through cables strictly in order of urgency (lowest rating first). At each step, they repair the next cable in the list *only if* it connects two currently isolated groups of data centers; otherwise they skip it and move on.

Unfortunately, urgency ratings have nothing to do with optimal repair costs — sometimes the most urgent cable is also the most expensive to fix. Calculate the total repair cost under this urgency-driven schedule, or determine that it is impossible to restore full connectivity.

Input Format

The first line contains two integers n and m ($1 \leq n \leq 10^5$, $1 \leq m \leq 2 \cdot 10^5$) — the number of data centers and cables.

Each of the next m lines contains three integers a , b , and c ($1 \leq a, b \leq n$, $1 \leq c \leq 10^9$) — a cable between data centers a and b with repair cost c .

All cables are bidirectional, each connects two different data centers, and all urgency ratings $p = c \cdot (a + b)$ are distinct.

Output Format

Print one integer — the total repair cost of the network built by the urgency-driven schedule.

If full connectivity cannot be restored even after processing every cable, print **IMPOSSIBLE**.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$
- $1 \leq c \leq 10^9$
- All urgency ratings $p = c \cdot (a + b)$ are distinct.

Examples

Sample Input	Sample Output
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4 5 1 2 10 3 4 2 2 4 6 1 3 8 2 3 3	15
4 3 1 2 4 2 3 7 1 3 2	IMPOSSIBLE

Notes

In the first sample, the urgency ratings are: $3 \leftrightarrow 4$ ($p = 14$), $2 \leftrightarrow 3$ ($p = 15$), $1 \leftrightarrow 2$ ($p = 30$), $1 \leftrightarrow 3$ ($p = 32$), $2 \leftrightarrow 4$ ($p = 36$). Processing in order: repair $3 \leftrightarrow 4$ (cost 2), repair $2 \leftrightarrow 3$ (cost 3), repair $1 \leftrightarrow 2$ (cost 10) — all four data centers are now connected. Total cost = $2 + 3 + 10 = 15$.

In the second sample, the urgency order is $1 \leftrightarrow 3$ ($p = 8$), $1 \leftrightarrow 2$ ($p = 12$), $2 \leftrightarrow 3$ ($p = 35$). The first two are repaired; the third is skipped as it would form a cycle. Data center 4 has no available cable, so full connectivity is impossible.

Marks Distribution

Task	Marks
Task A	60%
Task B	40%

Practice Problems

Problem	Links
Road Reparation	CSES: https://cses.fi/problemset/task/1675
Building Roads	CSES: https://cses.fi/problemset/task/1666
Edges in MST	Codeforces: https://codeforces.com/problemset/problem/160/D
Minimum Spanning Tree	Codeforces: https://codeforces.com/problemset/problem/17/B